Shresth Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.83
SHIVAM CONVENT NEW BYE PASS RD K BAGH PATNA BR (65414)	Class XII (CBSE)	2024	96.60%
DELHI PUBLIC SCHOOL LODHIPUR SHAHPUR PATNA BIHAR (65084)	Class X (CBSE)	2022	96.00%

WORK EXPERIENCE

Machine Learning Intern | BuildBay | [Dec '25] - [Feb '26]

- **Trained and optimized object detection models**, including Faster R-CNN and YOLO, to process and parse large-scale digital architectural datasets such as CubiCasa5k and ArchCAD-400k.
- **Extracted dimension endpoints from probability heatmaps** and implemented spatial logic to accurately pair them into precise, axis-aligned wall segments.
- **Integrated EasyOCR with a custom regression model** and utilized K-Means clustering algorithms to establish a highly accurate global pixel-to-meter ratio, successfully eliminating structural scaling errors.

PROJECTS

Video-Upload Backend Service | *Node.js, Express.js, MongoDB, JWT, Cloudinary*

- Architected a robust backend service for video handling, featuring secure user authentication, authorization, and session management via **JWT** and **bcrypt**.
- Engineered efficient media storage and processing pipelines by integrating **Multer** and **Cloudinary** for seamless file uploads and media management.
- Designed scalable database schemas using **MongoDB** and **Mongoose**, optimizing complex data retrieval with advanced aggregation pipelines and pagination.

High-Performance In-Memory Key-Value Store | *C++17, POSIX Sockets, Linux epoll, Docker, CMake*

- Engineered an in-memory Key-Value Database in C++17, implementing an **O(1) LRU cache** and a **Min-Heap priority queue** for active TTL key expiration.
- Architected a single-threaded, non-blocking TCP server using **Linux epoll**, bypassing mutex contention to multiplex concurrent clients and achieve **160,000+ peak local OPS.**
- Implemented an asynchronous Append-Only File (AOF) persistence layer, decoupling disk I/O from the main event loop to maximize write throughput, and containerized the deployment using **CMake** and **Docker.**

SKILLS

- **Programming Languages:** C++, C, JavaScript, Python, HTML, CSS
- **Tools:** Git/GitHub, Docker, Postman, SolidEdge
- **Development**: React, Node.js, Express.js
- **Databases:** MongoDB, PostgreSQL
- **Frameworks:** Pandas, NumPy, Seaborn, Matplotlib, OpenCV, PyTorch
- **Soft Skills:** Problem Solving, Leadership Skills, Team Collaboration, Effective Communication, Time Management

POSITIONS OF RESPONSIBILITY

Coordinator, Elan & nVision : *Annual Techno-cultural Fest, IIT Hyderabad*

- Direct artist management and mainstage production, handling everything from booking headliners to overseeing the audio-visual and stage setups.

Head, Elan & nVision : *Annual Techno-cultural Fest, IIT Hyderabad*

- Managed core on-ground logistics and venue infrastructure, coordinating with vendors to ensure smooth event execution, seating arrangements, and a reliable power supply.

RELEVANT COURSES

- **Computer Science:** Introduction To Programming, Data Structures & Applications
- **Mathematics:** Calculus, Probability, Linear Algebra, Transform Techniques, Statistics

SCHOLASTIC ACHIEVEMENTS

- Achieved a **maximum rating of 1393 on Codeforces**.
- Solved **600+ DSA questions** across multiple platforms (including LeetCode and Codeforces).